

The Running Recursion Summed Exactly

Telescoping product, second order and one-loop reading for Docs. 295 and 333

Johann Pascher
ORCID 0009-0000-6518-4064

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The Running Recursion Summed Exactly

Abstract

The recursion $\xi_{n+1} = \xi_n(1 - 100\xi_n)$ (Doc. 295, 333) is the bookkeeping of the fractal correction in the running case. Doc. 295 derives its continuum behaviour $\xi_n \approx 1/(100(n + 75))$ and the logarithmic defect $D(n) \approx \ln(1 + n/75)$. This document adds four statements. First, the product of all correction factors telescopes exactly to ξ_n/ξ_0 **[B]**. Second, a logarithmic second-order term improves the continuum approximation by a factor of 50 to 700 **[K]**. Third, the multiplicative and additive bookkeeping differ by a fixed amount $\psi'(75)/2$, and the bounded constant ≈ 0.0134 between mass and time defect found numerically in Doc. 295 is $\psi'(75) = \sum_{k \geq 0} (k + 75)^{-2}$ **[K]**. Fourth, the recursion has the form of a one-loop running, whose coefficient, however, is fixed only together with a step convention **[S]**.

Status markers

- [B]** Algebraically proved.
[K] Numerically confirmed (verification script, 120 digits).
[S] Structurally plausible, open.

1 Starting point

With $\xi_0 = 4/30000 = 1/7500$ one has $100\xi_0 = 1/75$ and the single step $K_{\text{frak}} = 1 - 100\xi_0 = 74/75$. In the frozen case this factor stays constant; in the running case (Case C, Doc. 333) it changes with every step according to

$$\xi_{n+1} = \xi_n (1 - 100\xi_n). \quad (1)$$

Doc. 295 shows that this turns into a logarithmic spiral with ratio e . The following sections compute how accurate the approximations used there are, and give in closed form the quantities that appear there only numerically or approximately.

2 The product telescopes

From Eq. (1) it follows that $1 - 100\xi_k = \xi_{k+1}/\xi_k$. The product of all correction factors therefore cancels down to the last and the first term:

$$\prod_{k=0}^{n-1} (1 - 100\xi_k) = \frac{\xi_n}{\xi_0}. \quad (2)$$

This is an exact identity **[B]**, confirmed in rational arithmetic for $n = 1$ to 12 **[K]**. The total accumulated fractal correction after n steps is thus simply the ratio of the last to the first ξ ; it need not be multiplied step by step. With the continuum form from Doc. 295 one obtains $\prod \approx 75/(n + 75)$, accurate to 0.007 % at $n = 10^5$ **[K]**.

3 Continuum at second order

With $u_n = 1/\xi_n$, Eq. (1) reads exactly $u_{n+1} = u_n + 100 + 100^2/(u_n - 100)$. Doc. 295 keeps only the step +100 and obtains $u_n \approx 100(n + 75)$. Including the next term $100^2/u_n \approx 100/(n + 75)$, it sums to a logarithm:

$$\frac{1}{\xi_n} \approx 100(n + 75) + 100 \ln \frac{n + 75}{75}. \quad (3)$$

Against the exact recursion (up to $n = 10^5$, 120 digits) **[K]**:

n	1st order (Doc. 295)	2nd order, Eq. (3)
1	0.018 %	0.0004 %
10	0.15 %	0.003 %
100	0.49 % (maximum)	0.005 %
10^4	0.049 %	0.00007 %
10^5	0.0072 %	0.000007 %

The approximation of Doc. 295 therefore never deviates by more than 0.5 % from the exact value; its statements (logarithmic spiral, ratio e , pole at $n = -75$) are unaffected. The second order is 50 to 700 times more accurate and is recommended wherever values of the running recursion are used numerically.

4 Multiplicative and additive: a fixed difference

Doc. 295 carries the accumulated defect additively, $D(n) = \sum_k 100 \xi_k$. The multiplicative bookkeeping is $-\ln \prod_k (1 - 100 \xi_k)$. Because $-\ln(1 - x) = x + x^2/2 + \dots$ it is always larger **[K]**, and the difference does not grow with n but tends to the sum of the second-order terms:

$$-\ln \prod_k (1 - 100 \xi_k) - D(n) \rightarrow \sum_{k \geq 0} \frac{1}{2(k+75)^2} = \frac{\psi'(75)}{2} = 0.00671 \quad \textbf{[K]}. \quad (4)$$

The two bookkeepings are therefore equivalent up to a fixed amount that can be given in closed form. The gap between the multiplicative form $(1 - \xi)^{100}$ and $74/75$, which Doc. 333 describes as the difference between frozen and running recursion, is a term of the same second-order kind.

The bounded constant from Doc. 295. Doc. 295 finds numerically that the accumulated defects of the mass side and the time side ($d_{\text{mass}} = 1/(1 - 100 \xi_n) - 1$, $d_{\text{time}} = 100 \xi_n$) differ only by a bounded constant ≈ 0.0134 . Their difference per step is $(100 \xi_n)^2/(1 - 100 \xi_n)$, and the sum tends to

$$\sum_{k \geq 0} \frac{1}{(k+75)^2} = \psi'(75) = 0.013423 \quad \textbf{[K]}, \quad (5)$$

to 6×10^{-5} relative. The constant is therefore not an accident of the simulation but the trigamma function at $75 = 1/(100 \xi_0)$, exactly twice the difference between multiplicative and additive bookkeeping.

5 The one-loop reading

In the continuum, Eq. (1) becomes $d\xi/dn = -100 \xi^2$ with the solution $\xi(n) = \xi_0/(1 + 100 n \xi_0)$. This is the form of a one-loop running of a coupling, as it holds for instance for α_s . The form is unique; the coefficient is not. If one step corresponds to a scale factor s , then $d\xi/d \ln \mu = -(100/\ln s) \xi^2$ **[K]**:

Step convention	$b = 100/\ln s$
Factor e (spiral ratio, Doc. 295)	100
Factor 2 (usual RG convention)	144
Factor $3/2$ ($q = 2/3$, Doc. 275)	247
Factor 10 (decade)	43

The identification with a β function is therefore a statement only once a step convention is fixed **[S]**. The spiral of Doc. 295 suggests the factor e , because it scales by e per precession turn; Doc. 275 works internally with $q = 2/3$. Which convention carries physical meaning is open.

6 Single step and power form

The single step $74/75$ can also be written in the power form $(D_f/3)^{D_f/2}$ (A040). With $D_f = 2.973$ it gives 0.9866508 against $74/75 = 0.9866667$, i.e. to 2×10^{-5} **[K]**. Exact agreement requires $D_f^{\text{eff}} = 2.973032$. Both forms are equivalent; the linear form $1 - 100 \xi_0$ is the one the recursion computes with.

7 Summary

Statement	Result	Status
Telescoping product, Eq. (2)	exactly ξ_n/ξ_0	[B]
Second order, Eq. (3)	50 to 700 times more accurate than Doc. 295	[K]
Multiplicative minus additive	$\psi'(75)/2 = 0.00671$	[K]
Bounded constant from Doc. 295	$\psi'(75) = 0.013423$	[K]
One-loop form	unique; coefficient depends on convention	[S]
Power form A040	$D_f^{\text{eff}} = 2.973032$	[K]

Doc. 295 remains valid in all its statements; its approximations are quantified here and extended by the second order. What remains open is which step convention underlies the one-loop reading physically, and, when several corrections run at the same time, whether they overlap. For the latter, the inclusion–exclusion rule of cluster expansions is the natural tool: the contribution of a level is its value minus what its sublevels already contain **[S]**.

Verification script. 2/python/Dok381_Skripte/pruef_381_rekursion_ansaetze.py (14/14): exact recursion in rational arithmetic and with 120 digits up to $n = 10^5$, telescoping product, both continuum orders, defect and product, constant from Doc. 295, power form, step conventions.